Python Dictionary

What is Python Dictionary?

- A data structure that allows retrieval by keyword
- Table of key-value pairs
- Commonly called *Associative array* (in general e.g. JavaScript, C++)
- Implemented using hashtable

Why the need to learn about dictionary?

Python Dictionary

Python dictionaries use the curly braces `{}`:

```
{key1:value1, key2:value2, ...}
```

```
e.g. {'Name':'James', 'Age':18, 'Gender':'M'}
```

Unique, Immutable

Key:	Value
'Name'	'James'
'Age'	18
'Gender'	'M'

One value per key, Any data type

Create Python Dictionary

```
>>> menu = {'soup': [('mushroom', 4),('tomato',3)],
'mains': ('chicken',10), 'dessert':{'weekdays':0,
 'weekends':('lava cake',7)}}
```

Key:	Value
'soup'	[('mushroom',4),('tomato',3)]
'mains'	('chicken',10)
'dessert'	{'weekdays':0, 'weekends':('lava cake',7)}

Create Python Dictionary

```
>>> {}  #empty dict using curly braces
{}
>>> dict()   #empty dict using built-in function
{}
```

Create Python Dictionary

Given a sequence (e.g. tuple, list) that contains key-value pair tuples:

```
>>> t = [('boys', 11), ('girls', 13)] #list containing tuples of key-value pairs)
>>> s =[('soup', [('mushroom', 4),('tomato',3)]),
('mains', ('chicken',10)), ('dessert',{'weekdays':0
, 'weekends':('lava cake',7)})]
```

Create Python Dictionary

```
>>> dict(t)
{'boys': 11, 'girls', 13}
>>> menu = dict(s)
>>> menu
{'soup': [('mushroom', 4),('tomato',3)],
'mains': ('chicken',10), 'dessert':{'weekdays':0,
  'weekends':('lava cake',7)}}
```

Create Python Dictionary

dict( ) applied to a sequence of key-value pairs

```
>>> dict([(1, 2), (2, 4), (3, 6)]) #list of kv tuples
{1:2, 2:4, 3:6}
```

Alternatively,

```
>>> dict(((1, 2), (2, 4), (3, 6))) #tuple of kv tuples
>>> dict([[1, 2], [2, 4], [3, 6]]) #list of kv list
```

Access value using key

```
>>> menu = {'soup': [('mushroom', 4),('tomato',3)],
'mains': ('chicken',10), 'dessert':{'weekdays':0,
  'weekends':('lava cake',7)}}


>>> menu['mains']
('chicken',10)
>>> menu['sides']
KeyError
```

Check whether entry exist

```
>>> menu = {'soup': [('mushroom', 4),('tomato',3)],
'mains': ('chicken',10), 'dessert':{'weekdays':0,
  'weekends':('lava cake',7)}}
##check for entry using in
>>> 'dessert' in menu  #only works for key
True
>>> 'chicken' in menu
False                    #since 'chicken' is not a key
```

Add/update an entry to dictionary

```
>>> menu['sides']= ('salad',4)  #add sides entry
>>> menu
{'soup': [('mushroom', 4),('tomato',3)],
'mains': ('chicken',10), 'dessert':{'weekdays':0,
  'weekends':('lava cake',7)}, 'sides':('salad',4)}
#Note that dictionary is mutable, and supports key-
  value assignment/re-assignment
```

Add/update an entry to dictionary

```
>>> menu['sides']= ('fries',4)    #update sides entry
>>> menu
{'soup': [('mushroom', 4),('tomato',3)],
'mains': ('chicken',10), 'dessert':{'weekdays':0,
  'weekends':('lava cake',7)}, 'sides':('fries',4)}
#sides value has been updated, in the same way sides
  was created
```

Delete entry

```
>>> del menu['soup']    #delete sides entry using key
>>> menu
{'mains': ('chicken',10), 'dessert':{'weekdays':0,
 'weekends':('lava cake',7)}, 'sides':('fries',4)}
>>> menu.clear() #delete all entries
```

List keys, values of dictionary

```
menu = {'mains': ('chicken',10),
  'dessert':{'weekdays':0, 'weekends':('lava
  cake',7)}, 'sides':('fries',4)}
>>> list(menu.keys())    #keys of menu
['mains', 'dessert', 'sides']
>>> list(menu.values())    #values of menu
[('chicken',10), {'weekdays':0, 'weekends':('lava
  cake',7)}, ('fries',4)]
```

Looping construct

```
>>> for key in menu:
        print menu[key]
('chicken',10)
{'weekdays':0, 'weekends':('lava cake',7)}
('fries',4)
>>> for key, value in menu.items():
        print (key, value)
'mains' ('chicken',10)
'dessert' {'weekdays':0, 'weekends':('lava cake',7)}
'sides'('fries',4)
```